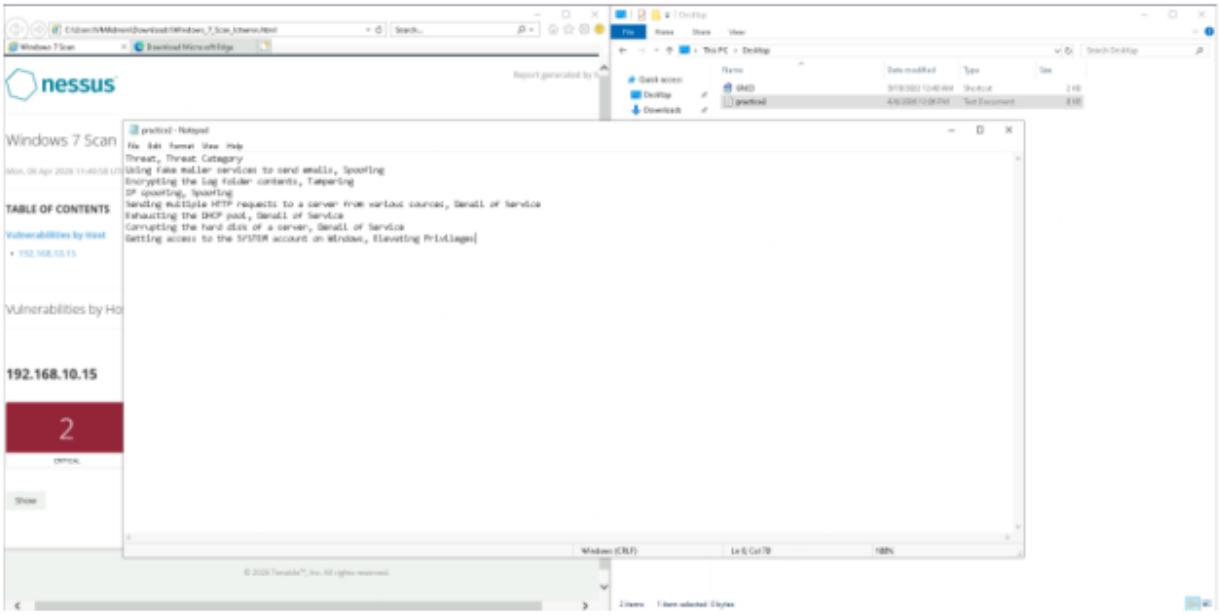
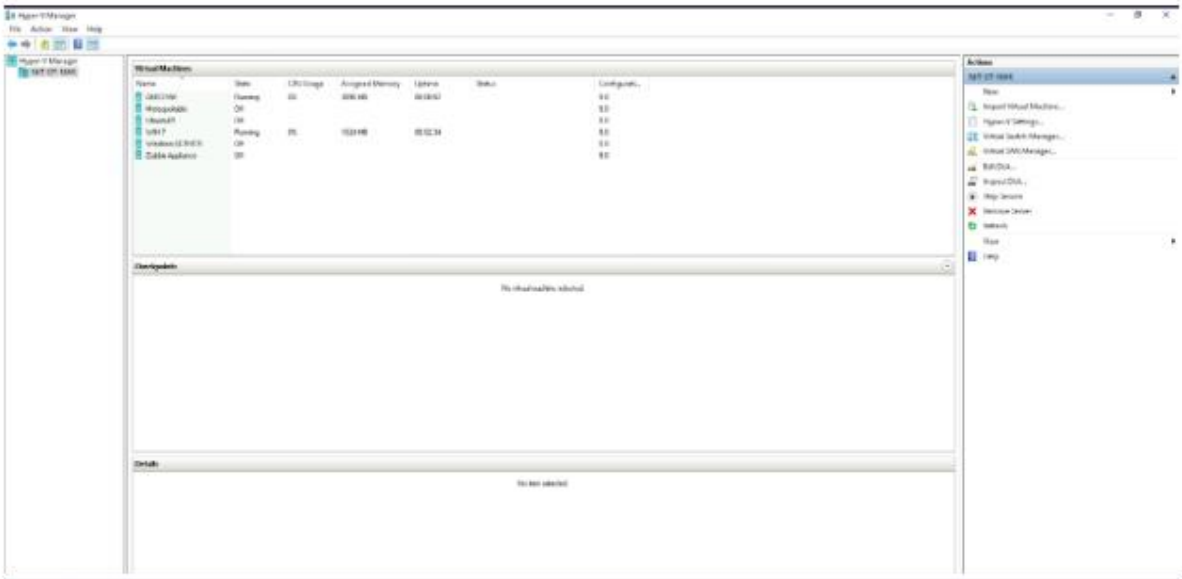
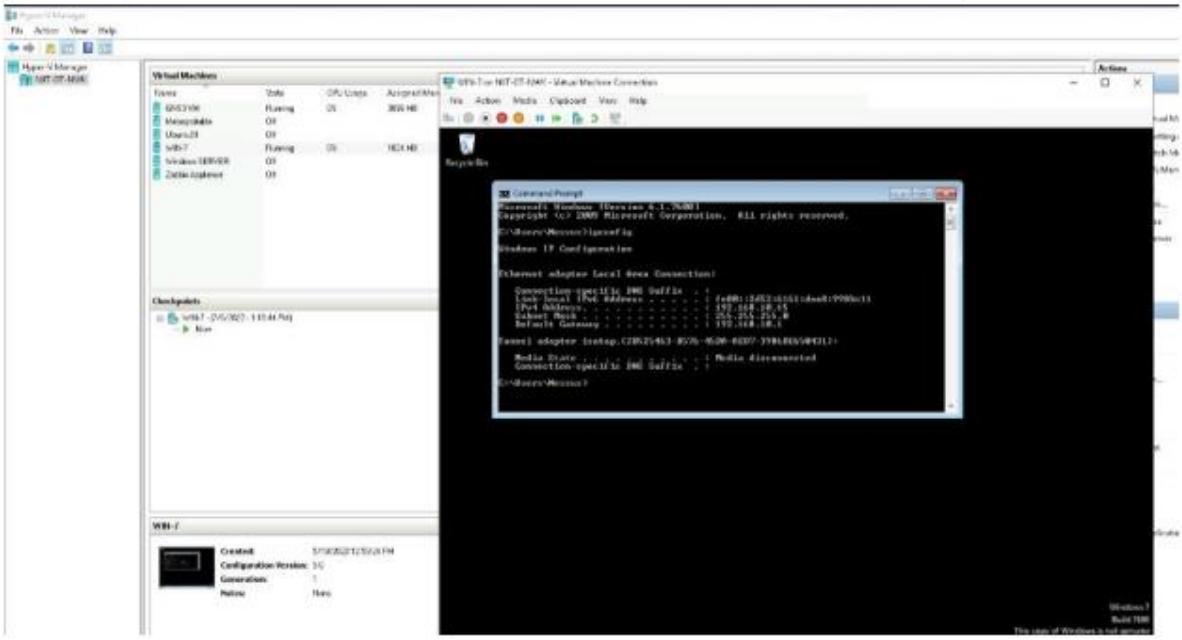
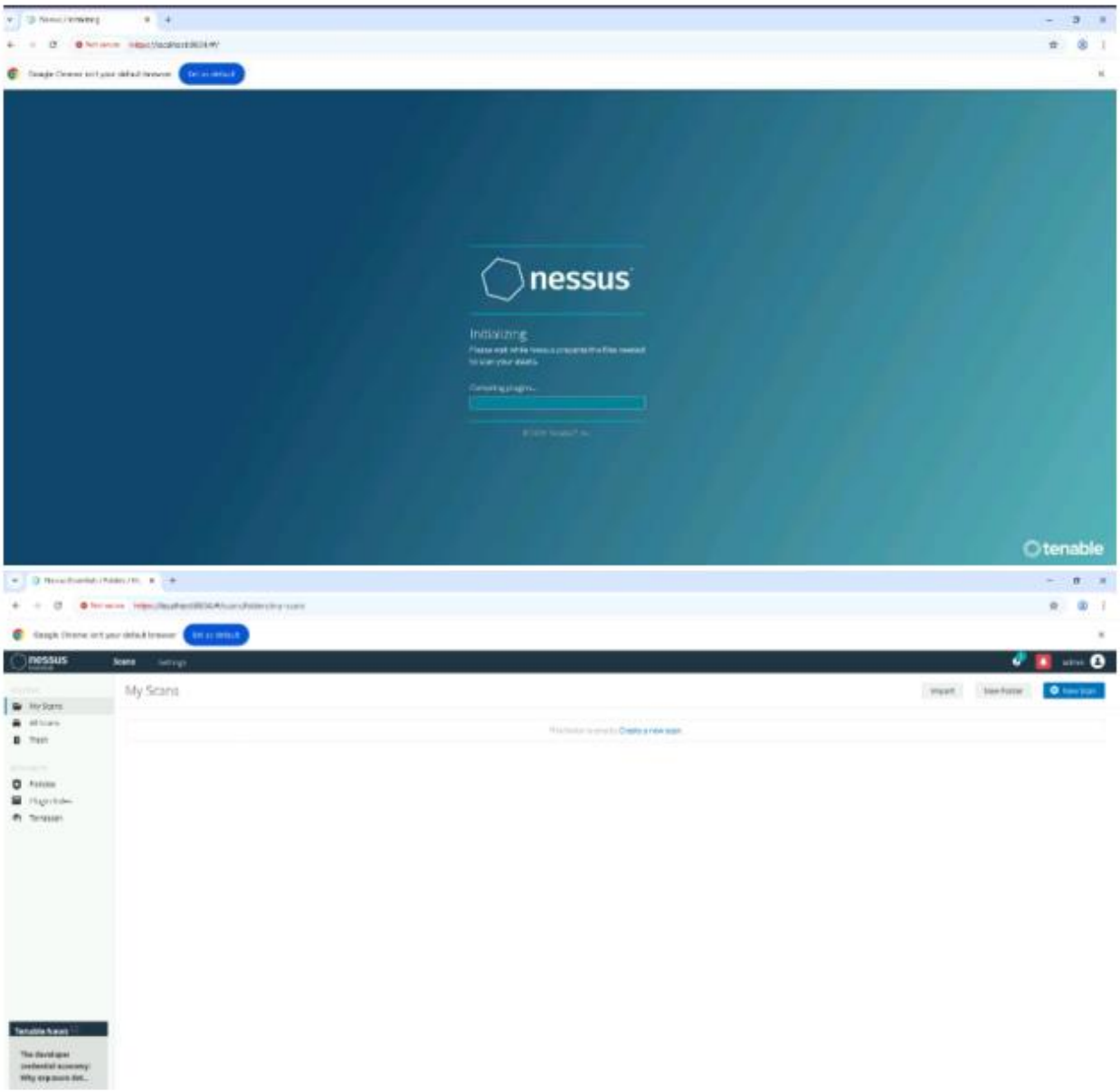
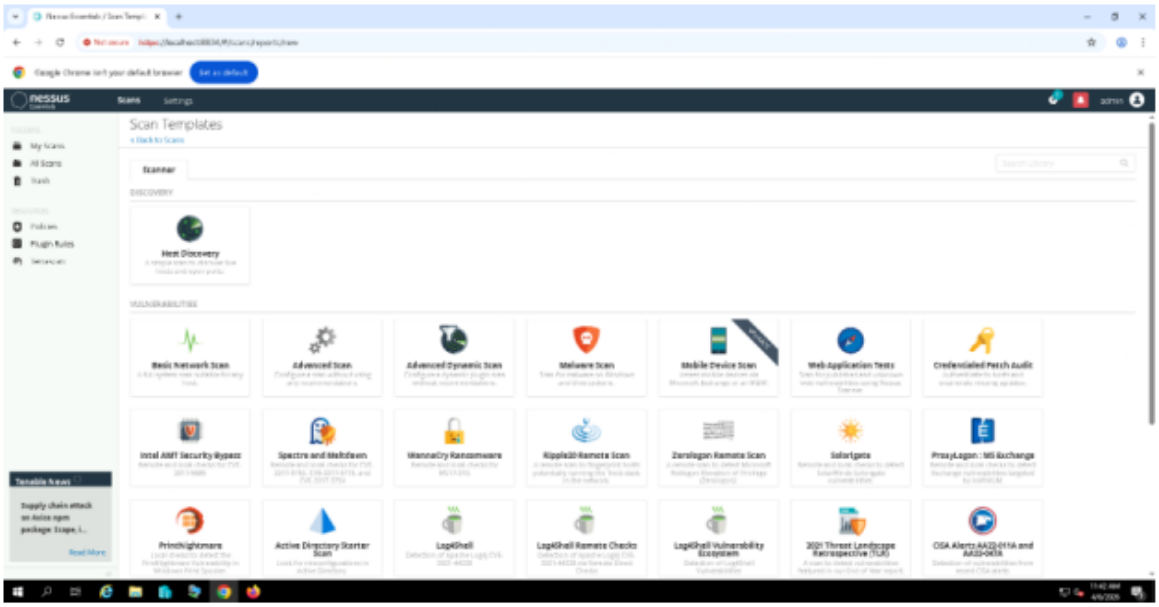
Scope 1:

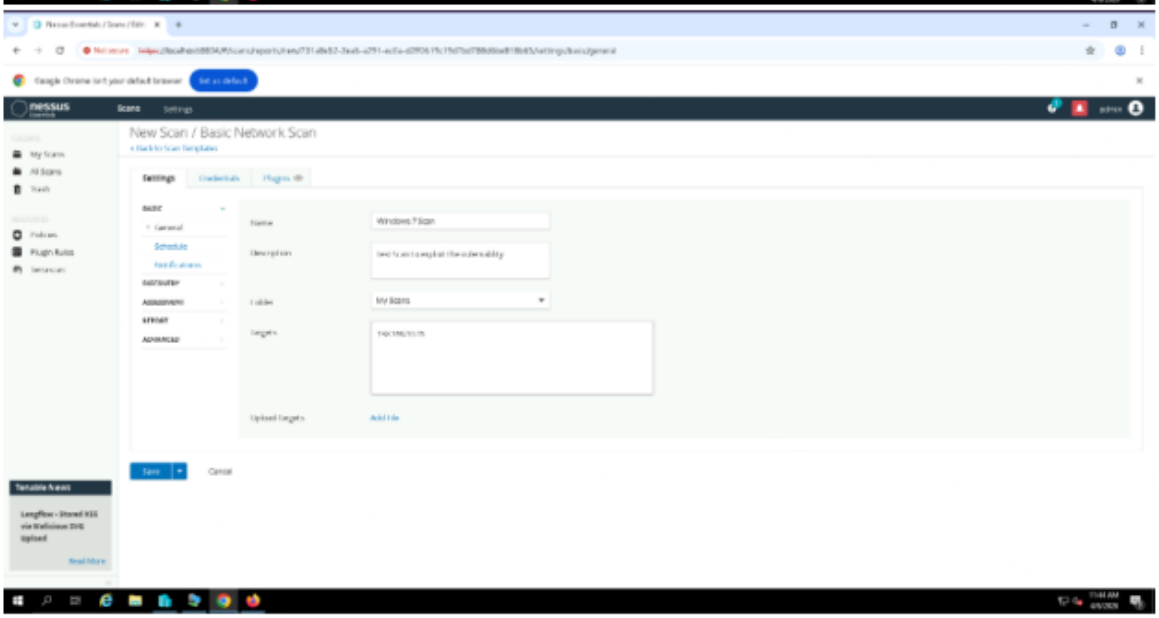

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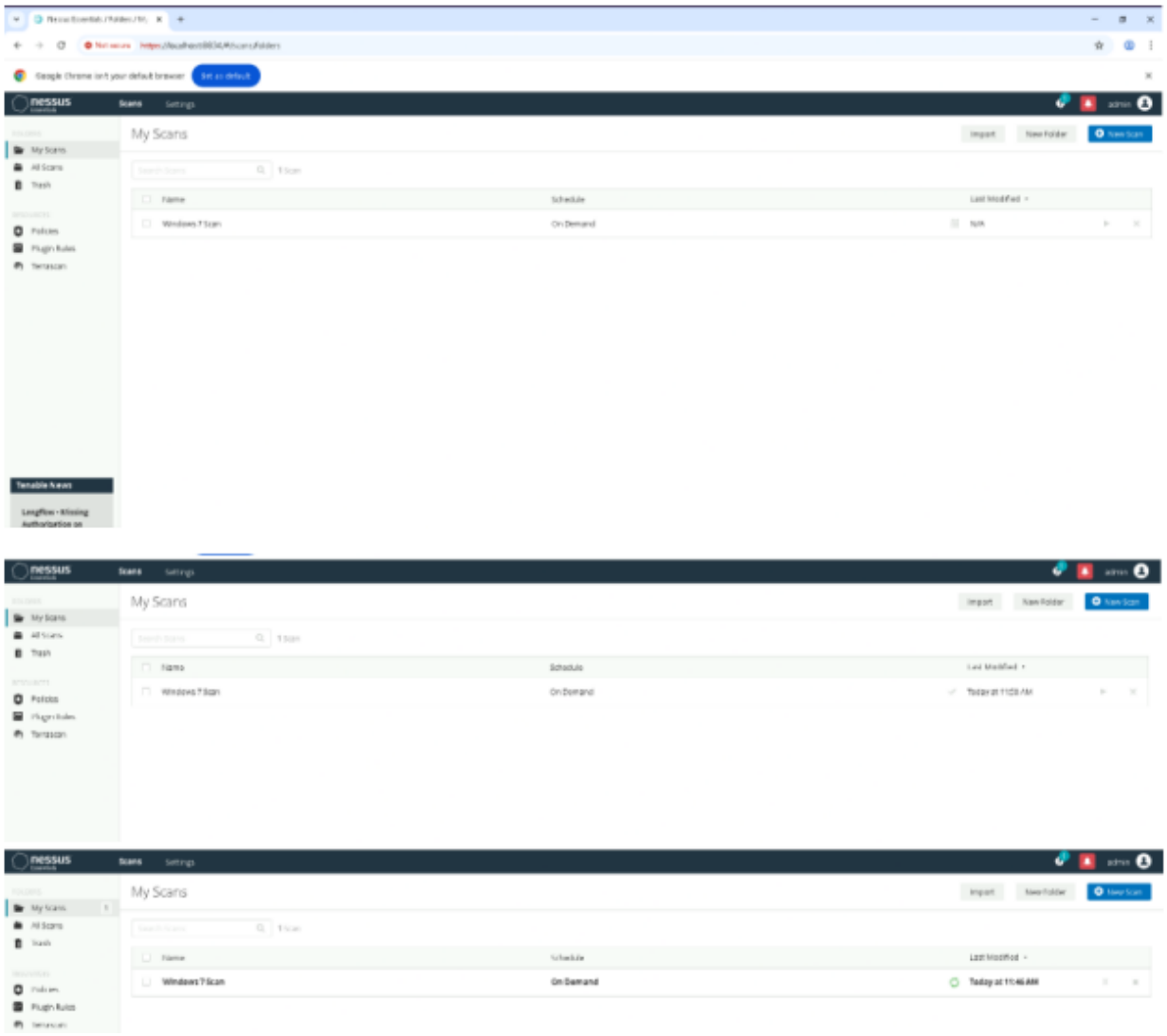


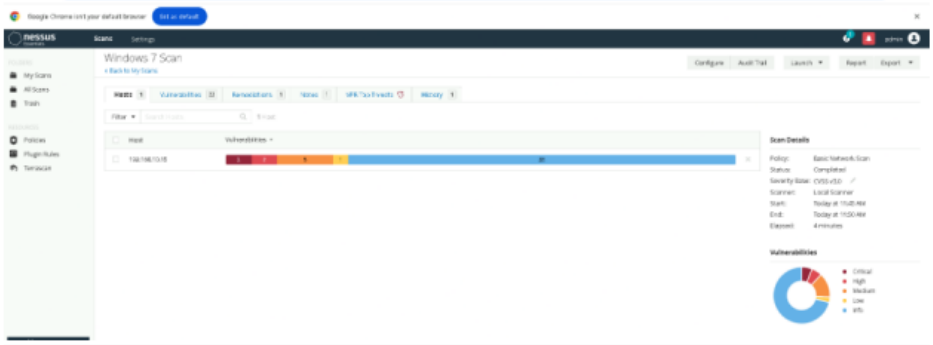


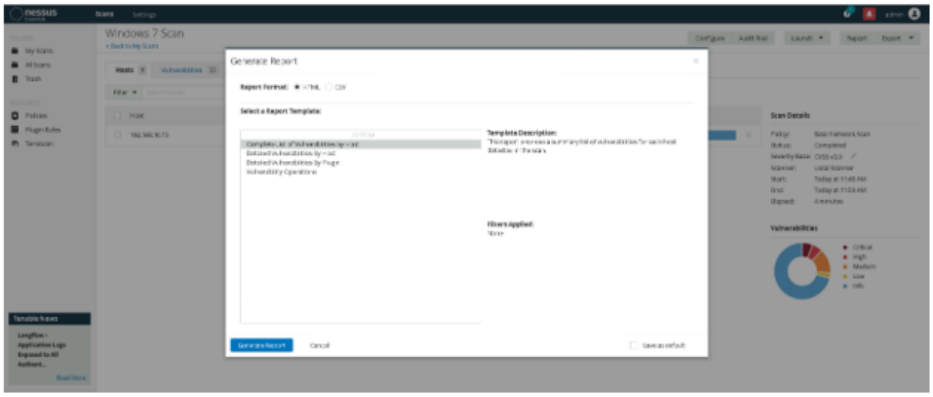

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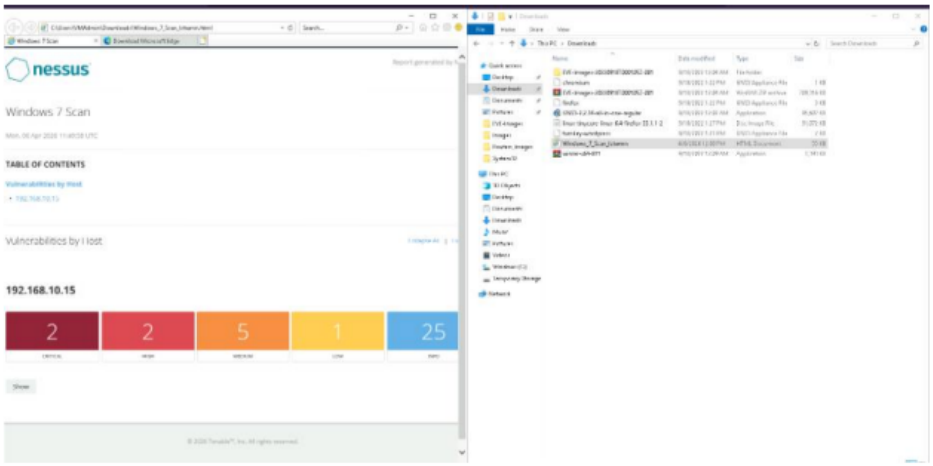

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Scope 2:

```
ubuntu@NIIT-DT-Linux:~/course6s2$ vagrant up
Bringing machine 'ldap.example.com' up with 'virtualbox' provider...
Bringing machine 'client.example.com' up with 'virtualbox' provider...
==> ldap.example.com: Importing base box 'ubuntu/bionic64'...
==> ldap.example.com: Matching MAC address for NAT networking...
==> ldap.example.com: Checking if box 'ubuntu/bionic64' version '20230607.0.5' i
s up to date...
==> ldap.example.com: Setting the name of the VM: course6s2_ldapexamplecom_17767
88069818_97337
==> ldap.example.com: Clearing any previously set network interfaces...
==> ldap.example.com: Preparing network interfaces based on configuration...
    ldap.example.com: Adapter 1: nat
    ldap.example.com: Adapter 2: hostonly
```

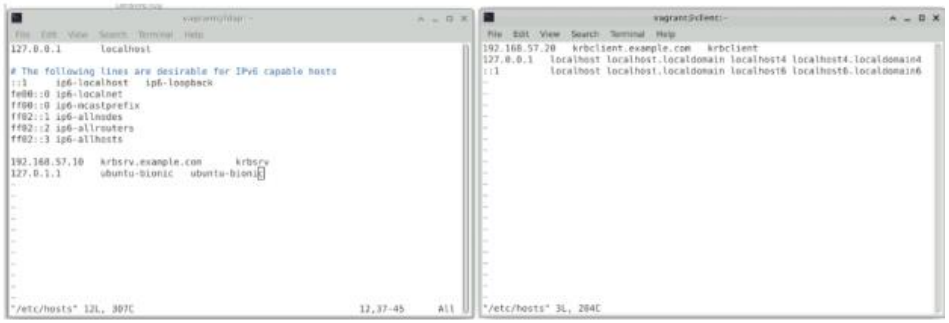


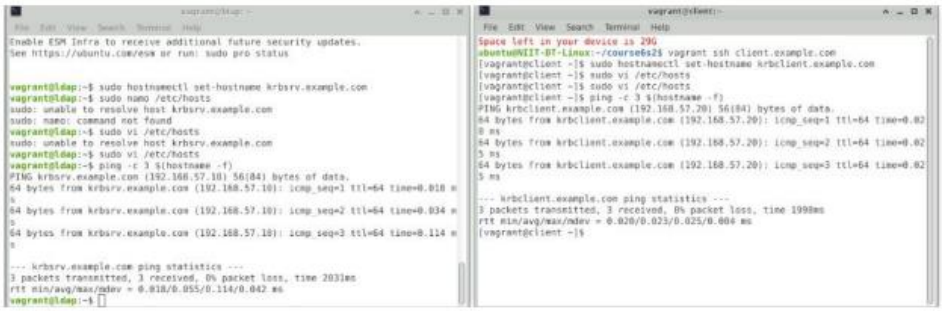

```
vagrant@ldap:~$ sudo hostnamectl set-hostname krbsrv.example.com
```

```
[vagrant@client ~]$ sudo hostnamectl set-hostname krbclient.example.com
```

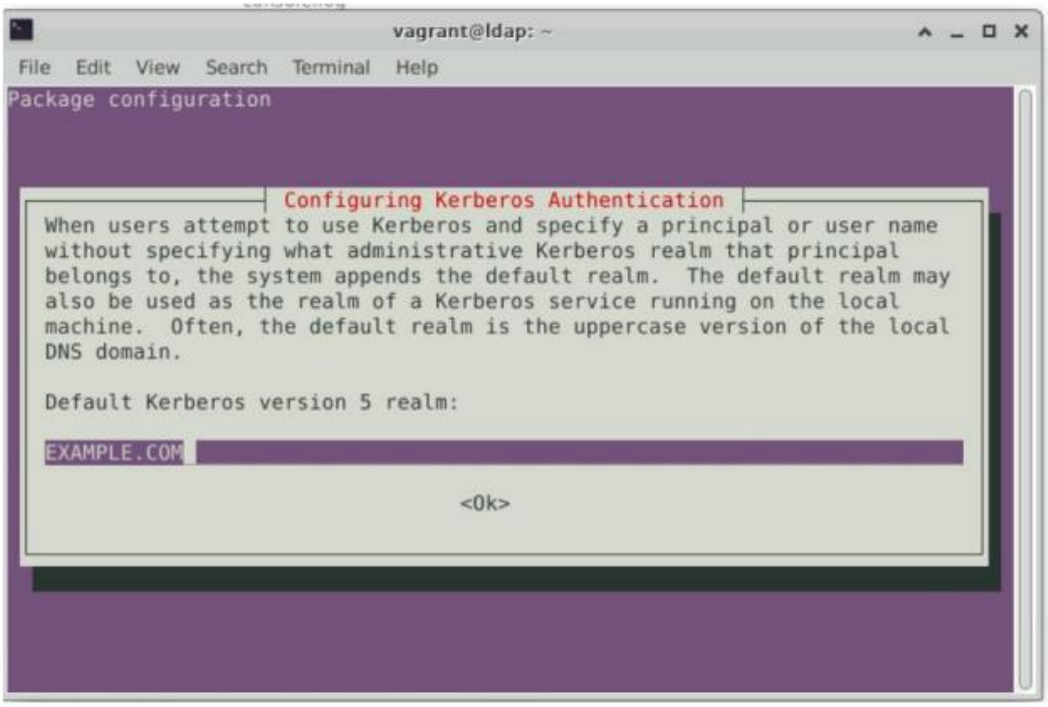
```
vagrant@ldap:~$ sudo vi /etc/hosts
```

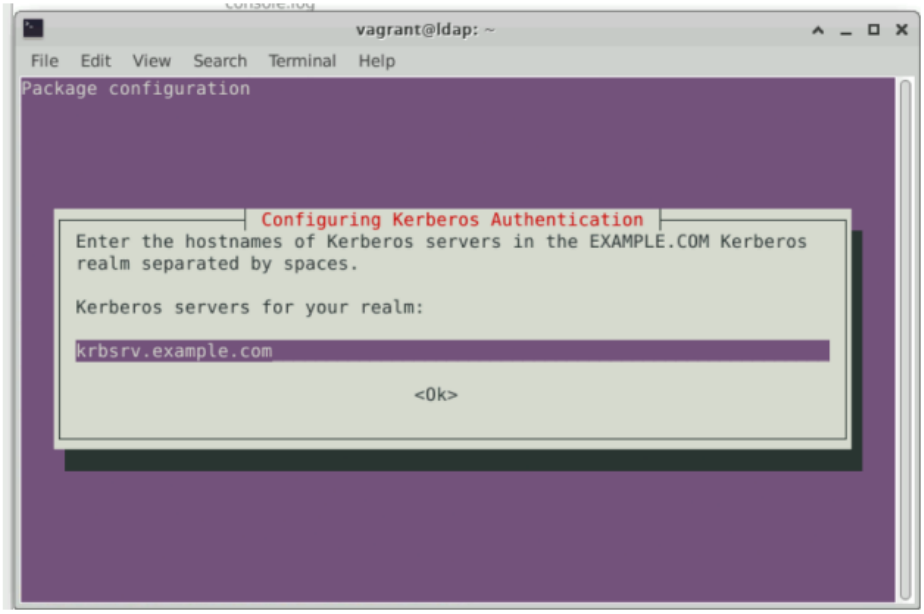
```
[vagrant@client ~]$ sudo vi /etc/hosts
```

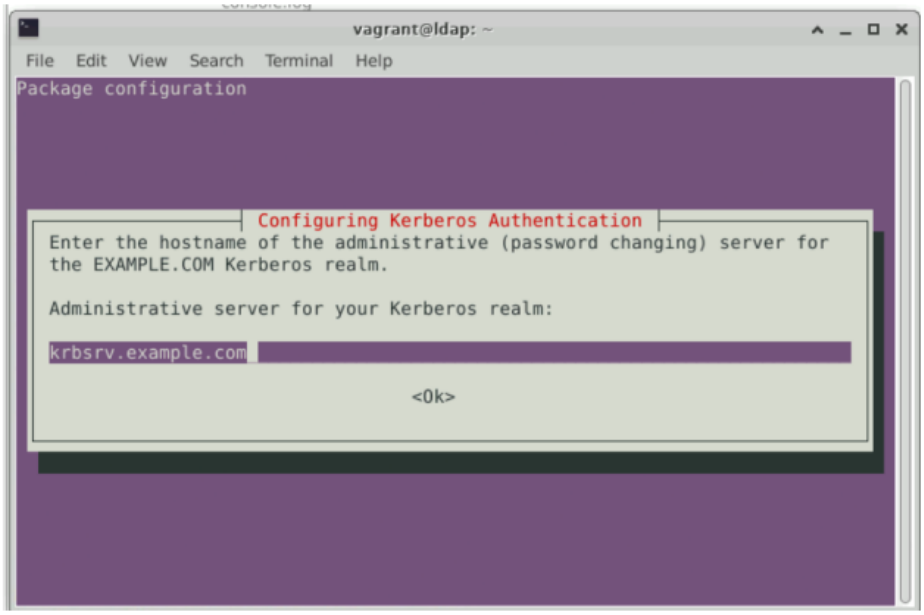


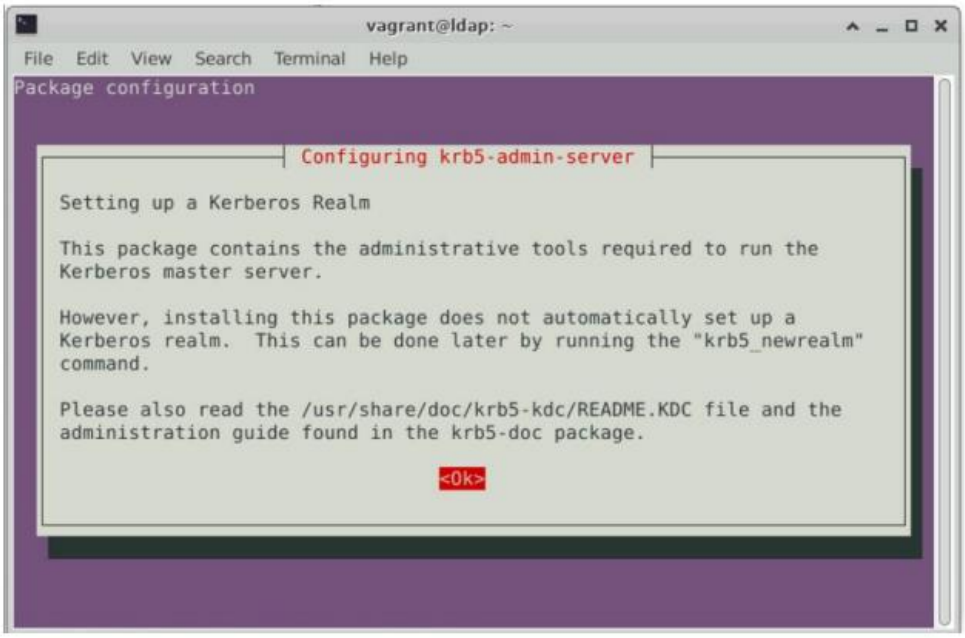


```
instead.
vagrant@ldap:~$ sudo apt install krb5-kdc krb5-admin-server krb5-config -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  krb5-user libgssrpc4 libkadm5clnt-mit11 libkadm5srv-mit11 libkdb5-9
  libverto-libevent1 libverto1
Suggested packages:
  krb5-kpropd krb5-kdc-ldap krb5-doc
The following NEW packages will be installed:
  krb5-admin-server krb5-config krb5-kdc krb5-user libgssrpc4
  libkadm5clnt-mit11 libkadm5srv-mit11 libkdb5-9 libverto-libevent1 libverto1
0 upgraded, 10 newly installed, 0 to remove and 17 not upgraded.
Need to get 585 kB of archives.
After this operation, 2277 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libgssrpc4 amd6
```




vagrant@ldap: ~
File Edit View Search Terminal Help
Package configuration
Configuring Kerberos Authentication
Enter the hostnames of Kerberos servers in the EXAMPLE.COM Kerberos realm separated by spaces.
Kerberos servers for your realm:
krbsrv.example.com
<Ok>


vagrant@ldap: ~
File Edit View Search Terminal Help
Package configuration
Configuring Kerberos Authentication
Enter the hostname of the administrative (password changing) server for the EXAMPLE.COM Kerberos realm.
Administrative server for your Kerberos realm:
krbsrv.example.com
<Ok>




```
vagrant@ldap:~$ sudo krb5_newrealm
This script should be run on the master KDC/admin server to initialize
a Kerberos realm.  It will ask you to type in a master key password.
This password will be used to generate a key that is stored in
/etc/krb5kdc/stash.  You should try to remember this password, but it
is much more important that it be a strong password than that it be
remembered.  However, if you lose the password and /etc/krb5kdc/stash,
you cannot decrypt your Kerberos database.
Loading random data
Initializing database '/var/lib/krb5kdc/principal' for realm 'EXAMPLE.COM',
master key name 'K/M@EXAMPLE.COM'
You will be prompted for the database Master Password.
It is important that you NOT FORGET this password.
Enter KDC database master key:
Re-enter KDC database master key to verify:


Now that your realm is set up you may wish to create an administrative
principal using the addprinc subcommand of the kadmin.local program.
Then, this principal can be added to /etc/krb5kdc/kadm5.acl so that
you can use the kadmin program on other computers.  Kerberos admin
principals usually belong to a single user and end in /admin.  For
example, if jruser is a Kerberos administrator, then in addition to
```

```
vagrant@ldap:~$ sudo kadmin.local
Authenticating as principal root/admin@EXAMPLE.COM with password.
kadmin.local:  addprinc root/admin
WARNING: no policy specified for root/admin@EXAMPLE.COM; defaulting to no policy
Enter password for principal "root/admin@EXAMPLE.COM":
Re-enter password for principal "root/admin@EXAMPLE.COM":
Principal "root/admin@EXAMPLE.COM" created.
kadmin.local:  addprinc -randkey host/krbsrv.example.com
WARNING: no policy specified for host/krbsrv.example.com@EXAMPLE.COM; defaulting
 to no policy
Principal "host/krbsrv.example.com@EXAMPLE.COM" created.
kadmin.local:  ktadd host/krbsrv.example.com
Entry for principal host/krbsrv.example.com with kvno 2, encryption type aes256-
cts-hmac-sha1-96 added to keytab FILE:/etc/krb5.keytab.
Entry for principal host/krbsrv.example.com with kvno 2, encryption type aes128-
cts-hmac-sha1-96 added to keytab FILE:/etc/krb5.keytab.
kadmin.local:  quit
vagrant@ldap:~$
```

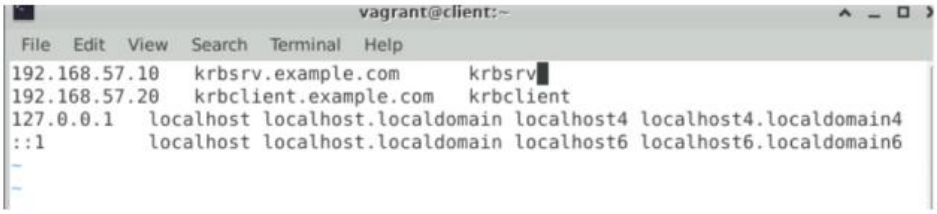
```
vagrant@ldap:~$ sudo vi /etc/krb5kdc/kadm5.acl
```

vagrant@ldap: ~

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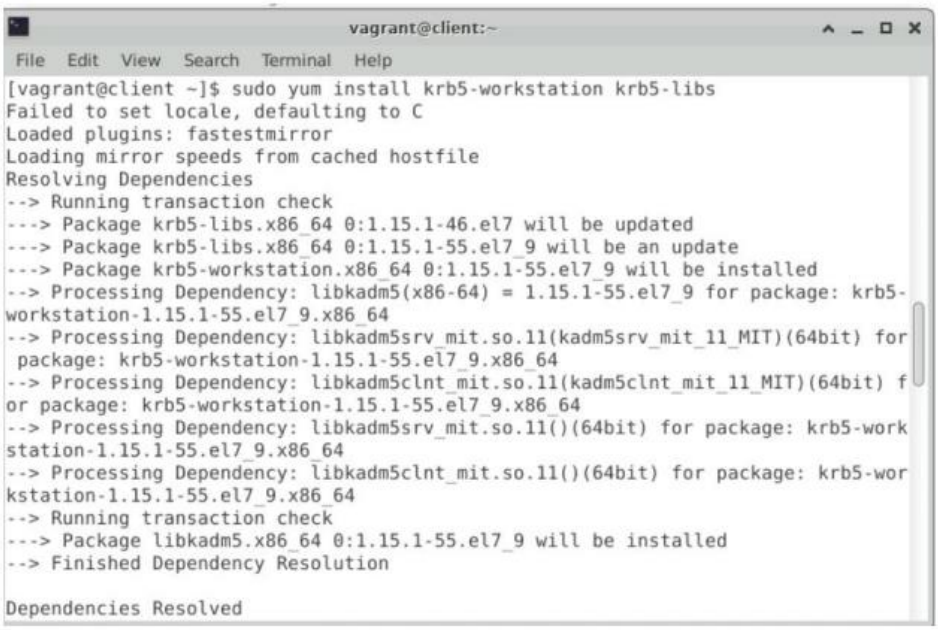
```
# This file Is the access control list for krb5 administration.
# When this file is edited run service krb5-admin-server restart to activate
# One common way to set up Kerberos administration is to allow any principal
# ending in /admin  is given full administrative rights.
# To enable this, uncomment the following line:
 root/admin *
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
-- INSERT --                                                    6,6           All
```

```
vagrant@ldap:~$ sudo systemctl restart krb5-admin-server.service
vagrant@ldap:~$
```

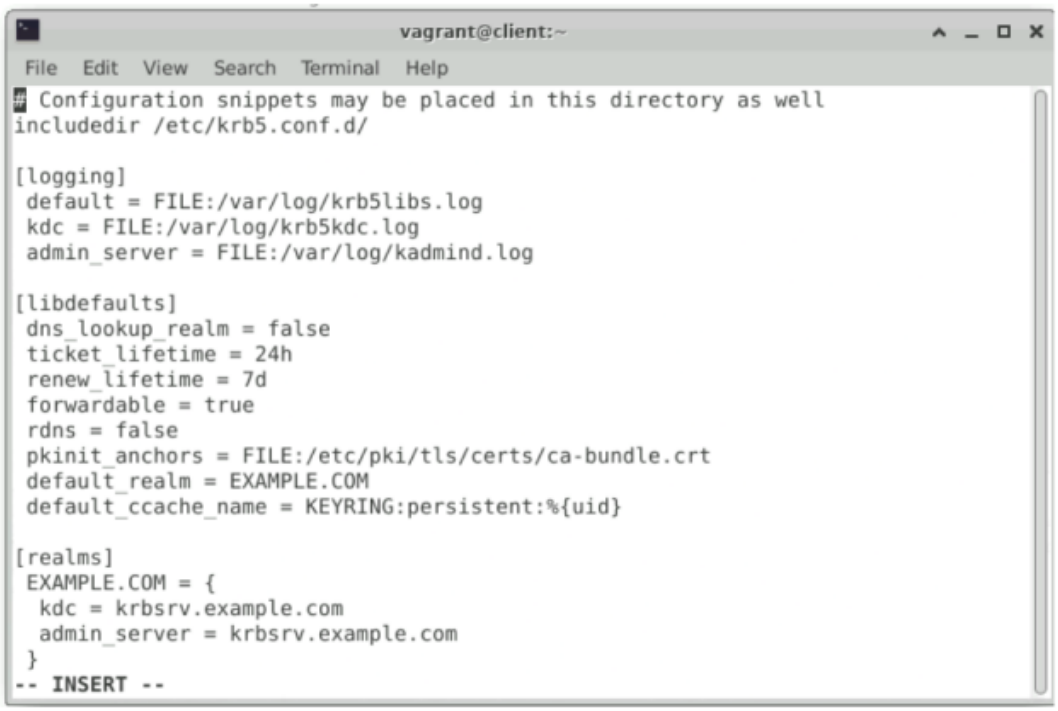


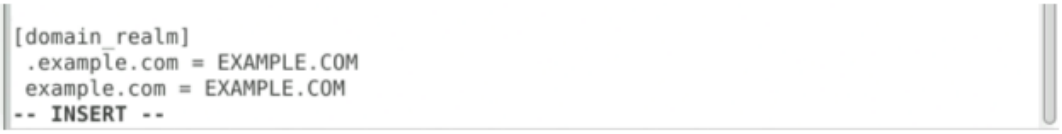

```
[vagrant@client ~]$ ping -c 3 krbsrv.example.com
PING krbsrv.example.com (192.168.57.10) 56(84) bytes of data.
64 bytes from krbsrv.example.com (192.168.57.10): icmp_seq=1 ttl=64 time=5.64 ms
64 bytes from krbsrv.example.com (192.168.57.10): icmp_seq=2 ttl=64 time=0.780 m
s
64 bytes from krbsrv.example.com (192.168.57.10): icmp_seq=3 ttl=64 time=0.635 m
s

--- krbsrv.example.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2001ms
rtt min/avg/max/mdev = 0.635/2.352/5.643/2.328 ms
[vagrant@client ~]$
```




```
Complete!
[vagrant@client ~]$ sudo vi /etc/krb5.conf
```


vagrant@client:~
File Edit View Search Terminal Help
Configuration snippets may be placed in this directory as well
includedir /etc/krb5.conf.d/
[logging]
default = FILE:/var/log/krb5libs.log
kdc = FILE:/var/log/krb5kdc.log
admin_server = FILE:/var/log/kadmind.log
[libdefaults]
dns_lookup_realm = false
ticket_lifetime = 24h
renew_lifetime = 7d
forwardable = true
rdns = false
pkinit_anchors = FILE:/etc/pki/tls/certs/ca-bundle.crt
default_realm = EXAMPLE.COM
default_ccache_name = KEYRING:persistent:%{uid}
[realms]
EXAMPLE.COM = {
kdc = krbsrv.example.com
admin_server = krbsrv.example.com
}
-- INSERT --


[domain_realm]
.example.com = EXAMPLE.COM
example.com = EXAMPLE.COM
-- INSERT --


[vagrant@client ~]$ sudo kadmin
Authenticating as principal root/admin@EXAMPLE.COM with password.
Password for root/admin@EXAMPLE.COM:
kadmin: addprinc -randkey host/krbclient.example.com
WARNING: no policy specified for host/krbclient.example.com@EXAMPLE.COM; defaulting to no policy
Principal "host/krbclient.example.com@EXAMPLE.COM" created.
kadmin: ktadd -k /etc/krb5.keytab host/krbclient.example.com
Entry for principal host/krbclient.example.com with kvno 2, encryption type aes256-cts-hmac-sha1-96 added to keytab WRFILE:/etc/krb5.keytab.
Entry for principal host/krbclient.example.com with kvno 2, encryption type aes128-cts-hmac-sha1-96 added to keytab WRFILE:/etc/krb5.keytab.
kadmin:

```
vagrant@ldap:~$ sudo useradd -m -s /bin/bash Patrick
vagrant@ldap:~$ sudo kadmin.local
Authenticating as principal root/admin@EXAMPLE.COM with password.
kadmin.local:  addprinc Patrick
WARNING: no policy specified for Patrick@EXAMPLE.COM; defaulting to no policy
Enter password for principal "Patrick@EXAMPLE.COM":
Re-enter password for principal "Patrick@EXAMPLE.COM":
Principal "Patrick@EXAMPLE.COM" created.
kadmin.local:  quit
vagrant@ldap:~$
```

```
vagrant@ldap:~$ sudo vi /etc/ssh/sshd_config
```

vagrant@ldap: ~

File Edit View Search Terminal Help

```
# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication no
#PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
ChallengeResponseAuthentication no

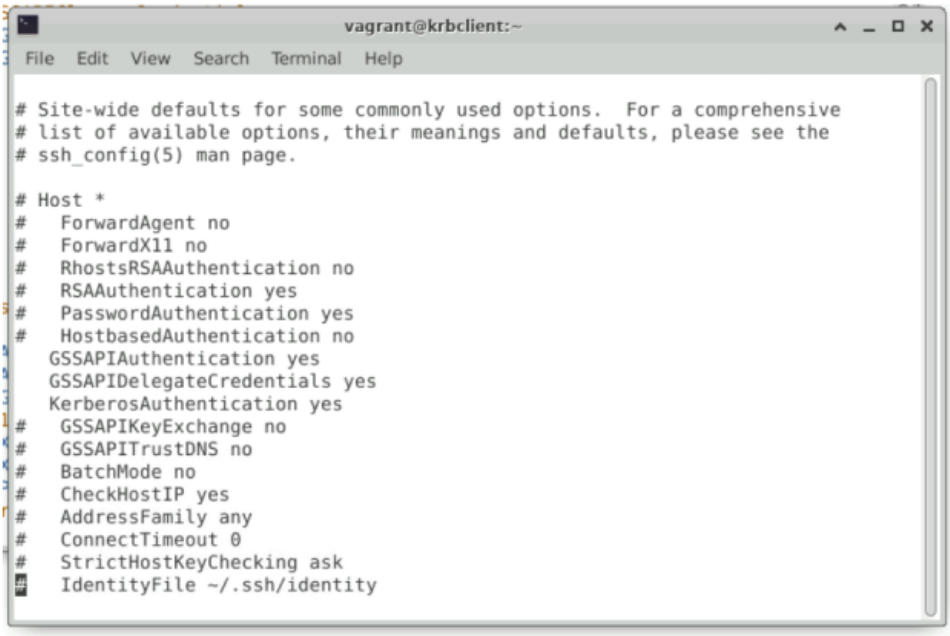
# Kerberos options
KerberosAuthentication yes
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

# GSSAPI options
GSSAPIAuthentication yes
GSSAPICleanupCredentials yes
#GSSAPIStrictAcceptorCheck yes
#GSSAPIKeyExchange no

# Set this to 'yes' to enable PAM authentication, account processing,
# and session processing. If this is enabled, PAM authentication will
-- INSERT --                                                    71,7          53%
```

```
vagrant@ldap:~$ sudo systemctl restart sshd
```

```
[vagrant@krbclient ~]$ sudo vi /etc/ssh/ssh_config
```




```
[vagrant@krbclient ~]$ sudo useradd -m -s /bin/bash Patrick
[vagrant@krbclient ~]$ sudo su
[root@krbclient vagrant]# kinit Patrick
Password for Patrick@EXAMPLE.COM:
[root@krbclient vagrant]# klist
Ticket cache: KEYRING:persistent:0:0
Default principal: Patrick@EXAMPLE.COM

Valid starting     Expires            Service principal
04/21/26 17:53:21  04/22/26 03:53:21  krbtgt/EXAMPLE.COM@EXAMPLE.COM
        renew until 04/28/26 17:53:17
[root@krbclient vagrant]# ssh Patrick@krbsrv.example.com
/etc/ssh/ssh_config line 29: Unsupported option "kerberosauthentication"
The authenticity of host 'krbsrv.example.com (192.168.57.10)' can't be establish
ed.
ECDSA key fingerprint is SHA256:j6stvrFjnkcZgwUzl37XD0SwwQLW1axRLtF0MMOUZZM.
ECDSA key fingerprint is MD5:53:47:c5:06:0d:29:b5:64:01:44:f8:05:ab:ba:73:a8.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added 'krbsrv.example.com,192.168.57.10' (ECDSA) to the lis
t of known hosts.
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-212-generic x86_64)
```

```
Patrick@krbsrv:~$ hostname
krbsrv.example.com
Patrick@krbsrv:~$ exit
logout
Connection to krbsrv.example.com closed.
[root@krbclient vagrant]# hostname
krbclient.example.com
[root@krbclient vagrant]# kinit Patrick
Password for Patrick@EXAMPLE.COM:

kinit: Password incorrect while getting initial credentials
[root@krbclient vagrant]#
[root@krbclient vagrant]#
[root@krbclient vagrant]# exit
exit
[vagrant@krbclient ~]$ sudo vi /etc/ssh/ssh_config
[vagrant@krbclient ~]$
```


vagrant@ldap: ~
File Edit View Search Terminal Help
Space left in your device is 31G
ubuntu@NIIT-DT-Linux:~/course6s2$ vagrant up
Bringing machine 'ldap.example.com' up with 'virtualbox' provider...
Bringing machine 'client.example.com' up with 'virtualbox' provider...
==> ldap.example.com: Importing base box 'ubuntu/bionic64'...
==> ldap.example.com: Matching MAC address for NAT networking...
==> ldap.example.com: Checking if box 'ubuntu/bionic64' version '20230607.0.5' i
s up to date...
==> ldap.example.com: Setting the name of the VM: course6s2_ldapexamplecom_17767
59302703_40931
==> ldap.example.com: Clearing any previously set network interfaces...
==> ldap.example.com: Preparing network interfaces based on configuration...
ldap.example.com: Adapter 1: nat
ldap.example.com: Adapter 2: hostonly
==> ldap.example.com: Forwarding ports...
ldap.example.com: 22 (guest) => 22010 (host) (adapter 1)
==> ldap.example.com: Running 'pre-boot' VM customizations...
==> ldap.example.com: Booting VM...
==> ldap.example.com: Waiting for machine to boot. This may take a few minutes..
.
ldap.example.com: SSH address: 127.0.0.1:22010
ldap.example.com: SSH username: vagrant
ldap.example.com: SSH auth method: private key
ldap.example.com: Warning: Remote connection disconnect. Retrying...

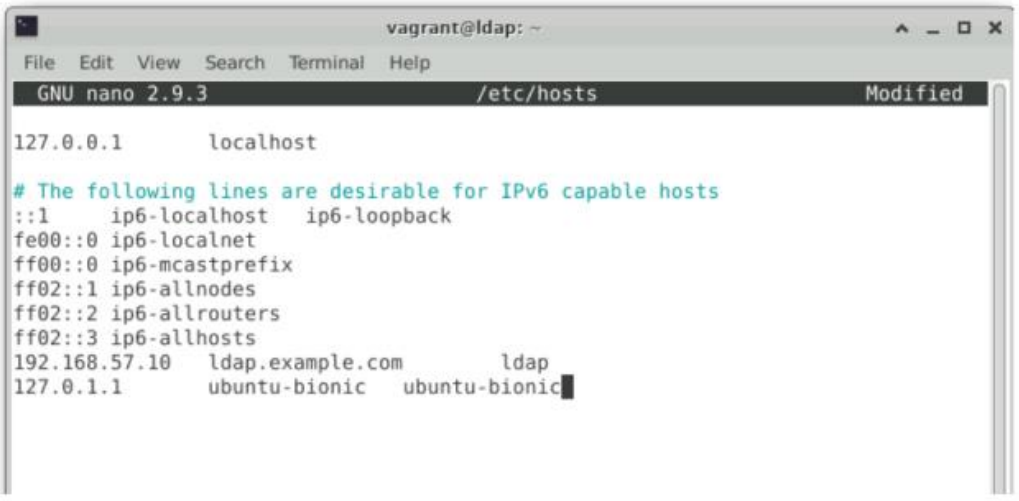
ubuntu@NIIT-DT-Linux:~/course6s2$ vagrant ssh ldap.example.com
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-212-generic x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/advantage
System information as of Tue Apr 21 08:18:12 UTC 2026
System load: 0.09 Processes: 91
Usage of /: 2.9% of 38.70GB Users logged in: 0
Memory usage: 12% IP address for enp0s3: 10.0.2.15
Swap usage: 0% IP address for enp0s8: 192.168.57.10
Expanded Security Maintenance for Infrastructure is not enabled.
0 updates can be applied immediately.
Enable ESM Infra to receive additional future security updates.

```
ubuntu@NIIT-DT-Linux:~/course6s2$ vagrant ssh client.example.com
[vagrant@client ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 52:54:00:4d:77:d3 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global noprefixroute dynamic eth0
       valid_lft 83536sec preferred_lft 83536sec
    inet6 fe80::5054:ff:fe4d:77d3/64 scope link
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:d4:32:33 brd ff:ff:ff:ff:ff:ff
    inet 192.168.57.20/24 brd 192.168.57.255 scope global noprefixroute eth1
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fed4:3233/64 scope link
       valid_lft forever preferred_lft forever
```

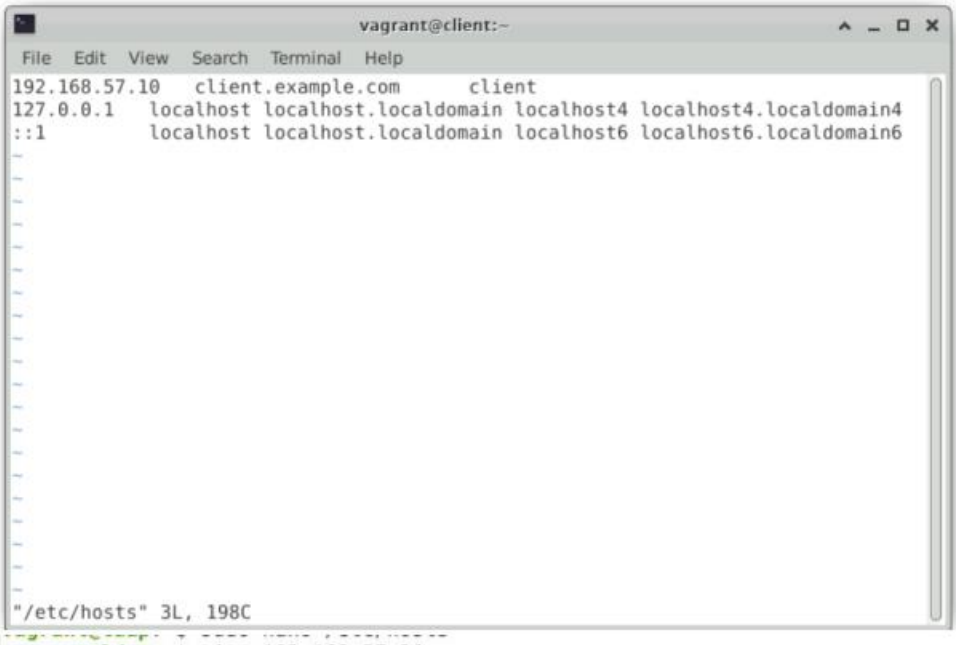
```
vagrant@ldap:~$ sudo hostnamectl set-hostname ldap.example.com
```

```
       valid_lft forever preferred_lft forever
[vagrant@client ~]$ sudo hostnamectl set-hostname client.example.com
```

```
vagrant@ldap:~$ sudo nano /etc/hosts
```




```
[vagrant@client ~]$ sudo vi /etc/hosts
```



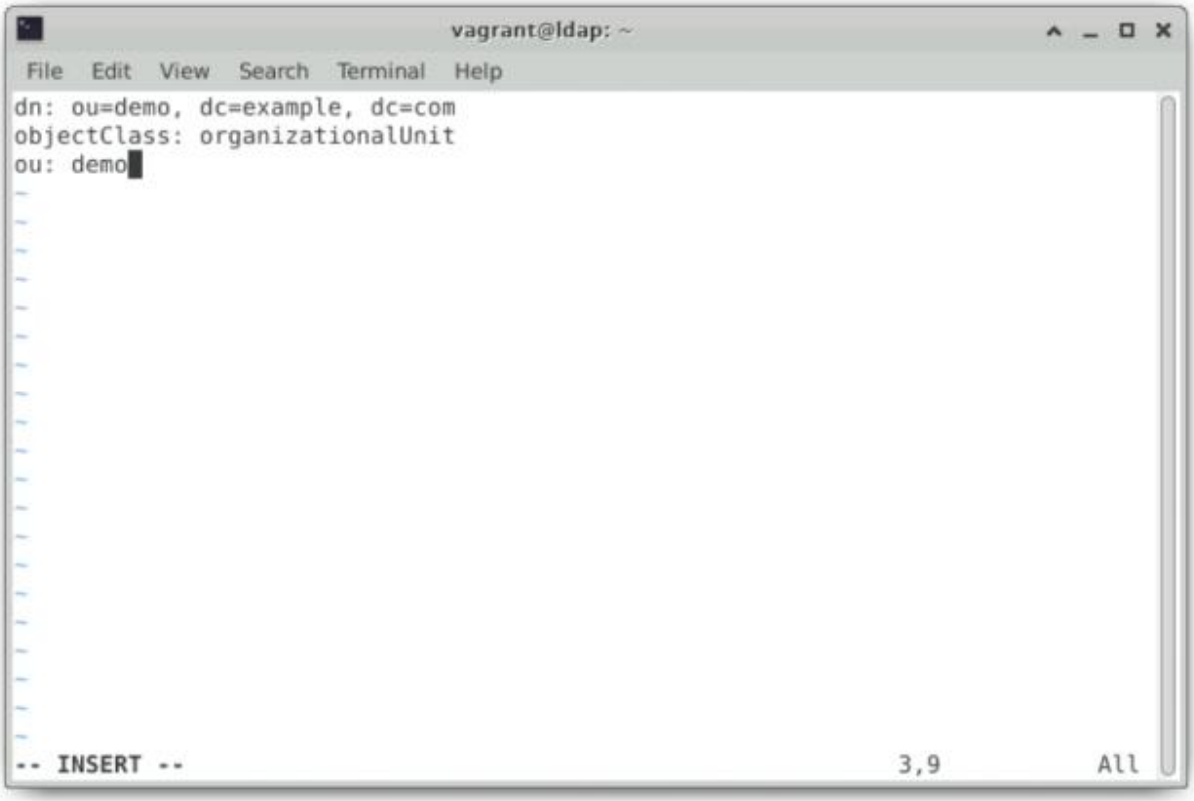

```
vagrant@ldap:~$ ping 192.168.57.20
PING 192.168.57.20 (192.168.57.20) 56(84) bytes of data.
64 bytes from 192.168.57.20: icmp_seq=1 ttl=64 time=2.92 ms
64 bytes from 192.168.57.20: icmp_seq=2 ttl=64 time=0.598 ms
64 bytes from 192.168.57.20: icmp_seq=3 ttl=64 time=0.652 ms
^C
--- 192.168.57.20 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 0.598/1.392/2.928/1.086 ms
```

```
[vagrant@client ~]$ ping 192.168.57.10
PING 192.168.57.10 (192.168.57.10) 56(84) bytes of data.
64 bytes from 192.168.57.10: icmp_seq=1 ttl=64 time=1.65 ms
64 bytes from 192.168.57.10: icmp_seq=2 ttl=64 time=1.86 ms
64 bytes from 192.168.57.10: icmp_seq=3 ttl=64 time=1.53 ms
64 bytes from 192.168.57.10: icmp_seq=4 ttl=64 time=1.88 ms
^C
--- 192.168.57.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.539/1.736/1.883/0.148 ms
```

```
vagrant@ldap:~$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu bionic-security InRelease [102 kB]
Hit:2 http://archive.ubuntu.com/ubuntu bionic InRelease
Get:3 http://archive.ubuntu.com/ubuntu bionic-updates InRelease [102 kB]
Get:4 http://security.ubuntu.com/ubuntu bionic-security/main amd64 Packages [271
7 kB]
Get:5 http://archive.ubuntu.com/ubuntu bionic-backports InRelease [102 kB]
Get:6 http://archive.ubuntu.com/ubuntu bionic/universe amd64 Packages [8570 kB]
Get:7 http://security.ubuntu.com/ubuntu bionic-security/main Translation-en [468
 kB]
Get:8 http://security.ubuntu.com/ubuntu bionic-security/restricted amd64 Package
```

```
vagrant@ldap:~$ sudo apt -y install slapd ldap-utils
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libltdl7 libodbc1
Suggested packages:
  libsasl2-modules-gssapi-mit | libsasl2-modules-gssapi-heimdal libmyodbc
  odbc-postgresql tdsodbc unixodbc-bin
The following NEW packages will be installed:
  ldap-utils libltdl7 libodbc1 slapd
0 upgraded, 4 newly installed, 0 to remove and 17 not upgraded.
Need to get 1731 kB of archives.
After this operation, 17.6 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu bionic/main amd64 libltdl7 amd64 2.4.6-2
[38.8 kB]
Get:2 http://archive.ubuntu.com/ubuntu bionic/main amd64 libodbc1 amd64 2.3.4-1.
```

```
vagrant@ldap:~$ sudo slapcat
dn: dc=example,dc=com
objectClass: top
objectClass: dcObject
objectClass: organization
o: example.com
dc: example
structuralObjectClass: organization
entryUUID: fc66dffa-d1af-1040-895e-f3b2b288d72e
creatorsName: cn=admin,dc=example,dc=com
createTimestamp: 20260421092656Z
entryCSN: 20260421092656.856373Z#000000#000#000000
modifiersName: cn=admin,dc=example,dc=com
modifyTimestamp: 20260421092656Z
```




```
vagrant@ldap:~$ sudo ldapadd -x -D cn=admin,dc=example,dc=com -W -f basedn.ldif
Enter LDAP Password:
adding new entry "ou=demo, dc=example, dc=com"

vagrant@ldap:~$ sudo slapcat
dn: dc=example,dc=com
objectClass: top
objectClass: dcObject
objectClass: organization
o: example.com
dc: example
structuralObjectClass: organization
```

```
entryUUID: fc6843fe-d1af-1040-895f-f3b2b288d72e
creatorsName: cn=admin,dc=example,dc=com
createTimestamp: 20260421092656Z
entryCSN: 20260421092656.865525Z#000000#000#000000
modifiersName: cn=admin,dc=example,dc=com
modifyTimestamp: 20260421092656Z

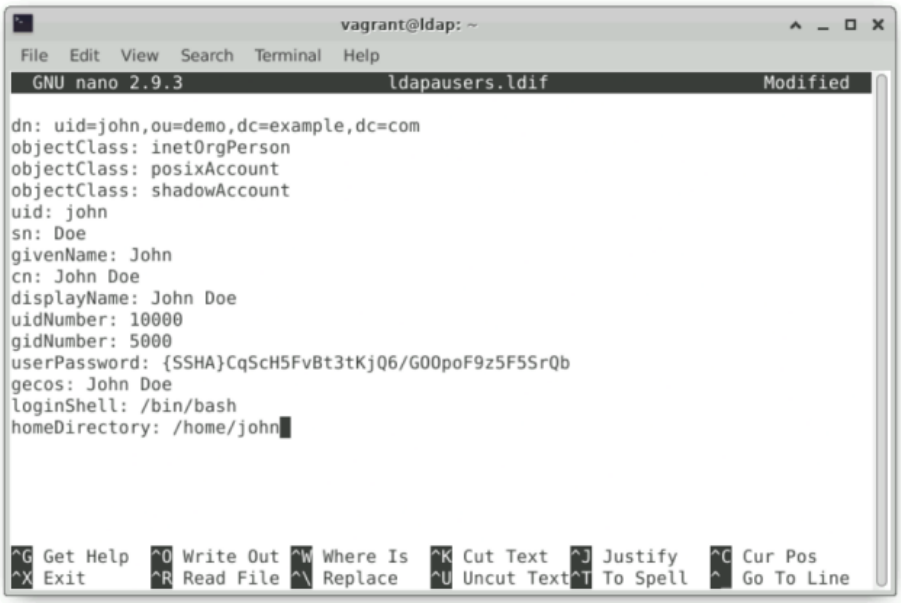
dn: ou=demo,dc=example,dc=com
objectClass: organizationalUnit
ou: demo
structuralObjectClass: organizationalUnit
entryUUID: e21aa2ba-d1b1-1040-8ba7-99cdf6c1e13f
creatorsName: cn=admin,dc=example,dc=com
createTimestamp: 20260421094031Z
entryCSN: 20260421094031.729473Z#000000#000#000000
modifiersName: cn=admin,dc=example,dc=com
modifyTimestamp: 20260421094031Z

vagrant@ldap:~$ sudo slappasswd
New password:
Re-enter new password:
{SSHA}CqScH5FvBt3tKjQ6/GOOpoF9z5F5SrQb
vagrant@ldap:~$ nano ldapausers.ldif
vagrant@ldap:~$ nano ldapausers.ldif
vagrant@ldap:~$ sudo ldapadd -x -D cn=admin,dc=example,dc=com -W -f ldapusers.ldif
ldapusers.ldif: No such file or directory
vagrant@ldap:~$ mv ldapausers.ldif ldapusers.ldif
vagrant@ldap:~$ sudo ldapadd -x -D cn=admin,dc=example,dc=com -W -f ldapusers.ldif
Enter LDAP Password:
adding new entry "uid=john,ou=demo,dc=example,dc=com"

vagrant@ldap:~$ ldapsearch -x -LL -b dc=example,dc=com '(uid=john)' cn gidNumberversion: 1

dn: uid=john,ou=demo,dc=example,dc=com
cn: John Doe
gidNumber: 5000

vagrant@ldap:~$ ldapsearch -x -LLL -b dc=example,dc=com '(ou=demo)' cn
dn: ou=demo,dc=example,dc=com
```




```
[vagrant@client ~]$ sudo yum install openldap openldap-clients nss-pam-ldapd-y
Failed to set locale, defaulting to C
Loaded plugins: fastestmirror
Loading mirror speeds from cached hostfile
No package nss-pam-ldapd-y available.
Resolving Dependencies
--> Running transaction check
---> Package openldap.x86_64 0:2.4.44-21.el7_6 will be updated
---> Package openldap.x86_64 0:2.4.44-25.el7_9 will be an update
---> Package openldap-clients.x86_64 0:2.4.44-25.el7_9 will be installed
--> Finished Dependency Resolution

Dependencies Resolved

================================================================================
 Package                Arch         Version                Repository     Size
================================================================================
Installing:
 openldap-clients       x86_64       2.4.44-25.el7_9        updates       191 k
Updating:
 openldap               x86_64       2.4.44-25.el7_9        updates       356 k

Transaction Summary
================================================================================
Install  1 Package
Upgrade  1 Package
```

```
[vagrant@client ~]$ sudo vi /usr/sbin/authconfig
[vagrant@client ~]$ sudo vi /etc/sysconfig/authconfig
```

```
[vagrant@client ~]$ sudo authconfig --enableldap --enableldapauth --ldapserver=192.168.57.10 --ldapbasedn="dc=example,dc=com" --enablemkhomedir --update
[vagrant@client ~]$ ls /usr/lib64/security/pam_ldap.so
/usr/lib64/security/pam_ldap.so
```

```
[vagrant@client ~]$ sudo vi /etc/openldap/ldap.conf
```

```
vagrant@client:~
File  Edit  View  Search  Terminal  Help
#
# LDAP Defaults
#

# See ldap.conf(5) for details
# This file should be world readable but not world writable.

#BASE   dc=example,dc=com
#URI    ldap://ldap.example.com ldap://ldap-master.example.com:666

#SIZELIMIT      12
#TIMELIMIT      15
#DEREF          never

TLS_CACERTDIR /etc/openldap/cacerts

# Turning this off breaks GSSAPI used with krb5 when rdns = false
SASL_NOCANON    on
URI ldap://192.168.57.10/
BASE dc=example,dc=com
```

```
[vagrant@client ~]$ ldapsearch -x
# extended LDIF
#

# john, demo, example.com
dn: uid=john,ou=demo,dc=example,dc=com
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: john
sn: Doe
givenName: John
cn: John Doe
displayName: John Doe
uidNumber: 10000
gidNumber: 5000
gecos: John Doe
loginShell: /bin/bash
homeDirectory: /home/john
```

```
[vagrant@client ~]$ sudo yum install ssd oddjob-mkhomedir.x86_64 -y
Failed to set locale, defaulting to C
Loaded plugins: fastestmirror
Loading mirror speeds from cached hostfile
No package ssd available.
Resolving Dependencies
--> Running transaction check
---> Package oddjob-mkhomedir.x86_64 0:0.31.5-4.el7 will be installed
--> Processing Dependency: oddjob = 0.31.5-4.el7 for package: oddjob-mkhomedir-0.31.5-4.el7.x86_64
--> Processing Dependency: psmisc for package: oddjob-mkhomedir-0.31.5-4.el7.x86_64
--> Running transaction check
---> Package oddjob.x86_64 0:0.31.5-4.el7 will be installed
---> Package psmisc.x86_64 0:22.20-17.el7 will be installed
--> Finished Dependency Resolution

Dependencies Resolved
```

```
[vagrant@client ~]$ sudo vi /etc/pam.d/system-auth
[vagrant@client ~]$
```

Applications Terminal course6s2 - File Manager

vagrant@client:~

File Edit View Search Terminal Help

```
#%PAM-1.0
# This file is auto-generated.
# User changes will be destroyed the next time authconfig is run.
auth        required      pam_env.so
auth        required      pam_faildelay.so delay=2000000
auth        sufficient    pam_unix.so nullok try_first_pass
auth        requisite     pam_succeed_if.so uid >= 1000 quiet_success
auth        sufficient    pam_ldap.so use_first_pass
auth        required      pam_deny.so

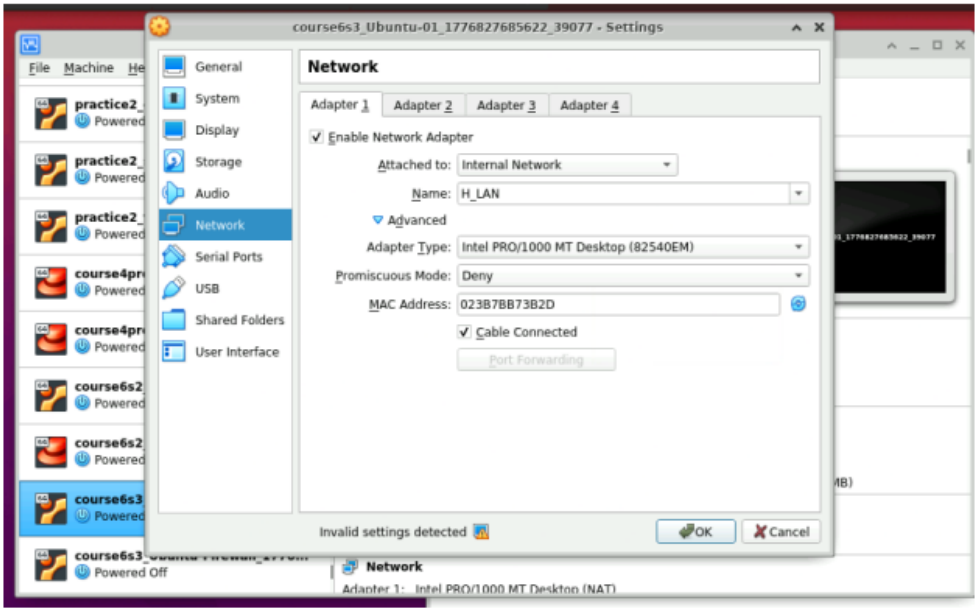
account     required      pam_unix.so broken_shadow
account     sufficient    pam_localuser.so
account     sufficient    pam_succeed_if.so uid < 1000 quiet
account     [default=bad success=ok user_unknown=ignore] pam_ldap.so
account     required      pam_permit.so

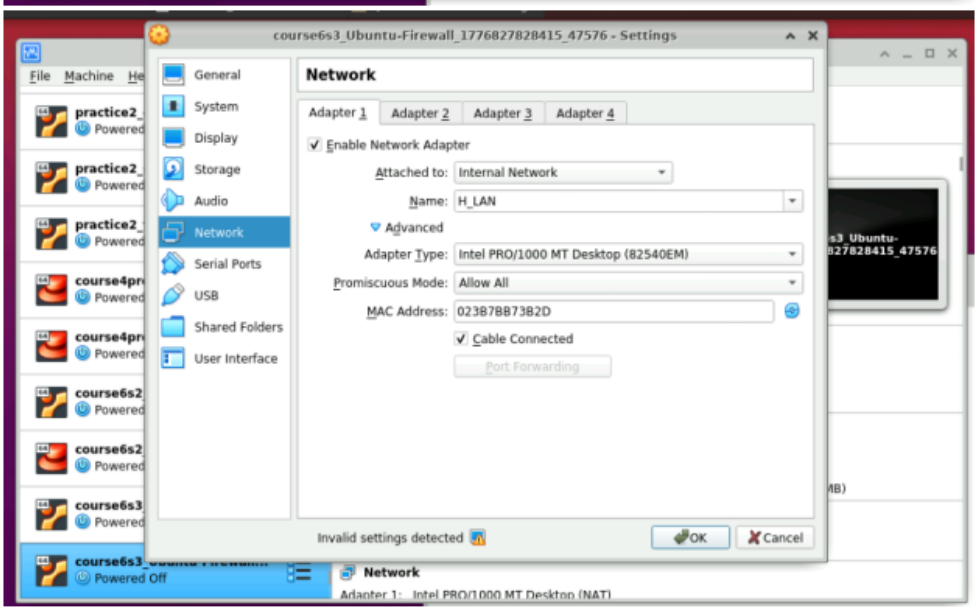
password    requisite     pam_pwquality.so try_first_pass local_users_only retry=3 authtok_type=
password    sufficient    pam_unix.so md5 shadow nullok try_first_pass use_authtok
password    sufficient    pam_ldap.so use_authtok
password    required      pam_deny.so

session     optional      pam_keyinit.so revoke
session     required      pam_oddjob_mkhomedir.so
-session     optional      pam_systemd.so
session     optional      pam_oddjob_mkhomedir.so umask=0077
session     [success=1 default=ignore] pam_succeed_if.so service in crond quiet use_uid
session     required      pam_oddjob_mkhomedir.so
session     optional      pam_ldap.so
~
~
~
~
~
```

```
[vagrant@client ~]$ sudo systemctl enable oddjobd.service --now
Created symlink from /etc/systemd/system/multi-user.target.wants/oddjobd.service to /usr/lib/systemd/system/oddjobd.service.
[vagrant@client ~]$ sudo systemctl start oddjobd.service --now
[vagrant@client ~]$ sudo authconfig --enablemkhomedir --updateall
[vagrant@client ~]$ sudo systemctl restart sshd.service
[vagrant@client ~]$ su - root
Password:
[root@client ~]# su - john
Creating home directory for john.
/usr/bin/id: cannot find name for group ID 5000
[john@client ~]$
```

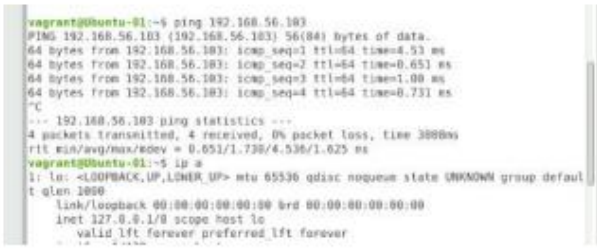
Scope 3:

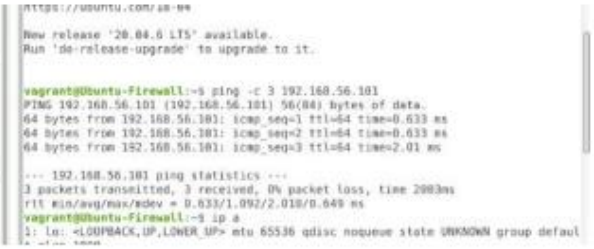

course6s3_Ubuntu-01_1776827685622_39077 - Settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to: Internal Network
Name: H_LAN
Advanced
Adapter Type: Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode: Deny
MAC Address: 023B7BB73B2D
Cable Connected
Port Forwarding
Invalid settings detected
OK
Cancel
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)

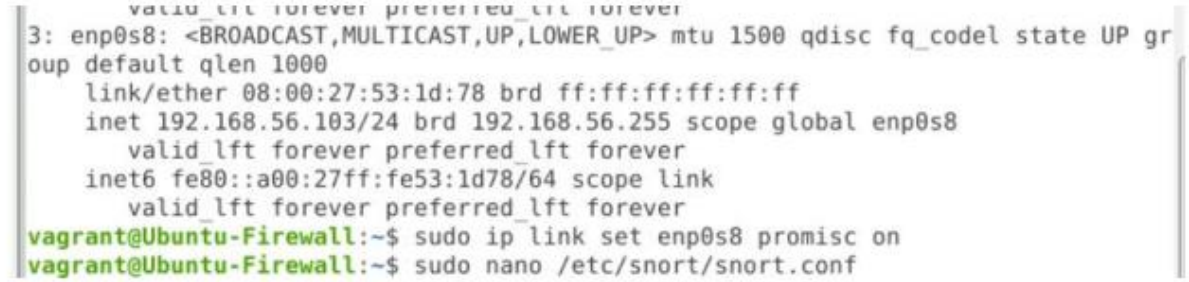

course6s3_Ubuntu-Firewall_1776827828415_47576 - Settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to: Internal Network
Name: H_LAN
Advanced
Adapter Type: Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode: Allow All
MAC Address: 023B7BB73B2D
Cable Connected
Port Forwarding
Invalid settings detected
OK
Cancel
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)









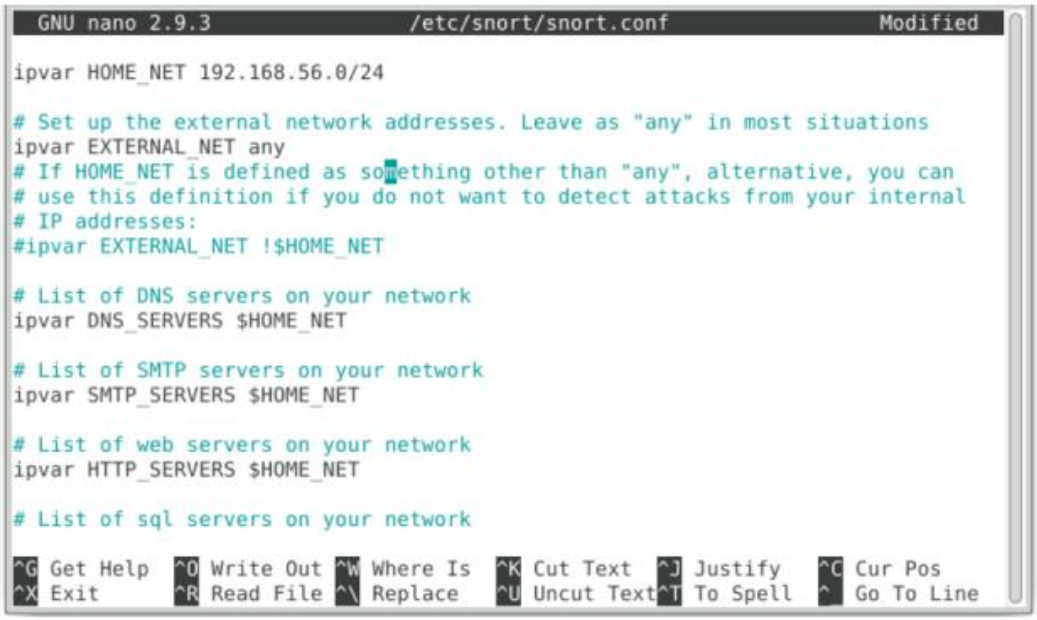




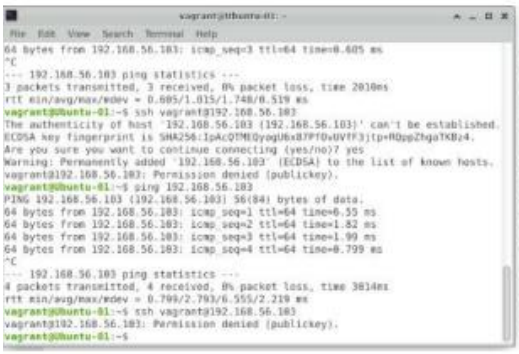



```
alert icmp any any <> 192.168.56.103 any (msg:"ICMP Found"; sid:1000000001;)

alert tcp any any <> 192.168.56.103 22 (msg:"SSH Found"; flow:to_server; flags:S; sid:1000000002;)
```


GNU nano 2.9.3 /etc/snort/snort.conf Modified
ipvar HOME_NET 192.168.56.0/24
Set up the external network addresses. Leave as "any" in most situations
ipvar EXTERNAL_NET any
If HOME_NET is defined as something other than "any", alternative, you can
use this definition if you do not want to detect attacks from your internal
IP addresses:
#ipvar EXTERNAL_NET !$HOME_NET
List of DNS servers on your network
ipvar DNS_SERVERS $HOME_NET
List of SMTP servers on your network
ipvar SMTP_SERVERS $HOME_NET
List of web servers on your network
ipvar HTTP_SERVERS $HOME_NET
List of sql servers on your network
^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^X Exit ^R Read File ^\ Replace ^U Uncut Text ^T To Spell ^_ Go To Line

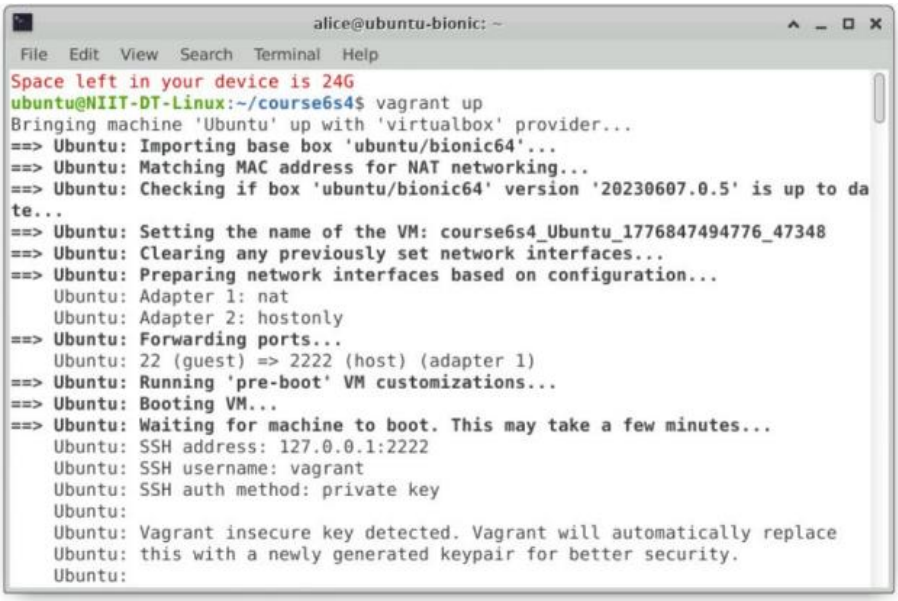



vagrant@ubuntu-01: ~
File Edit View Search Terminal Help
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.605 ms
^C
--- 192.168.56.103 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2010ms
rtt min/avg/max/mdev = 0.605/1.015/1.748/0.519 ms
vagrant@ubuntu-01:~$ ssh vagrant@192.168.56.103
The authenticity of host '192.168.56.103 (192.168.56.103)' can't be established.
ECDSA key fingerprint is SHA256:IpAcQTMEQyagU6x87PfOuVYF3jtp+RQppZhgaTKBz4.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.168.56.103' (ECDSA) to the list of known hosts.
vagrant@192.168.56.103: Permission denied (publickey).
vagrant@ubuntu-01:~$ ping 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.55 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=1.82 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=1.99 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.799 ms
^C
--- 192.168.56.103 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3814ms
rtt min/avg/max/mdev = 0.799/2.793/6.555/2.210 ms
vagrant@ubuntu-01:~$ ssh vagrant@192.168.56.103
vagrant@192.168.56.103: Permission denied (publickey).
vagrant@ubuntu-01:~$


Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Commencing packet processing (pid=5052)
04/22-08:16:37.987691 [**] [1:366:7] ICMP PING *NIX [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.56.101 -> 192.168.56.103
04/22-08:16:37.987691 [**] [1:384:5] ICMP PING [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.56.101 -> 192.168.56.103
04/22-08:16:37.987717 [**] [1:408:5] ICMP Echo Reply [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.56.103 -> 192.168.56.101
04/22-08:16:38.987836 [**] [1:366:7] ICMP PING *NIX [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.56.101 -> 192.168.56.103
04/22-08:16:38.987836 [**] [1:384:5] ICMP PING [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.56.101 -> 192.168.56.103
04/22-08:16:38.987866 [**] [1:408:5] ICMP Echo Reply [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.56.103 -> 192.168.56.101
04/22-08:16:39.993369 [**] [1:366:7] ICMP PING *NIX [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.56.101 -> 192.168.56.103
04/22-08:16:39.993369 [**] [1:384:5] ICMP PING [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.56.101 -> 192.168.56.103
04/22-08:16:39.993654 [**] [1:408:5] ICMP Echo Reply [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.56.103 -> 192.168.56.101
04/22-08:16:40.997957 [**] [1:366:7] ICMP PING *NIX [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.56.101 -> 192.168.56.103
04/22-08:16:40.997957 [**] [1:384:5] ICMP PING [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.56.101 -> 192.168.56.103


vagrant@Ubuntu-Firewall:~$ sudo ls /var/log/snort
snort.log.1776845791 snort.log.1776846555
vagrant@Ubuntu-Firewall:~$
vagrant@Ubuntu-Firewall:~$ sudo cat /var/log/snort/alert
cat: /var/log/snort/alert: No such file or directory
vagrant@Ubuntu-Firewall:~$ sudo cat /var/log/snort/snort.log1776846555
cat: /var/log/snort/snort.log1776846555: No such file or directory
vagrant@Ubuntu-Firewall:~$ sudo cat /var/log/snort/snort.log.1776846555
vagrant@Ubuntu-Firewall:~$

Scope 4:




```
ubuntu@NIIT-DT-Linux:~/course6s4$ vagrant ssh Ubuntu
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-212-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage
```

```
vagrant@ubuntu-bionic:~$ sudo adduser alice
Adding user `alice' ...
Adding new group `alice' (1002) ...
Adding new user `alice' (1002) with group `alice' ...
Creating home directory `/home/alice' ...
Copying files from `/etc/skel' ...
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
Changing the user information for alice
Enter the new value, or press ENTER for the default
        Full Name []: alice
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
vagrant@ubuntu-bionic:~$ sudo usermod -a -G sudo alice
```

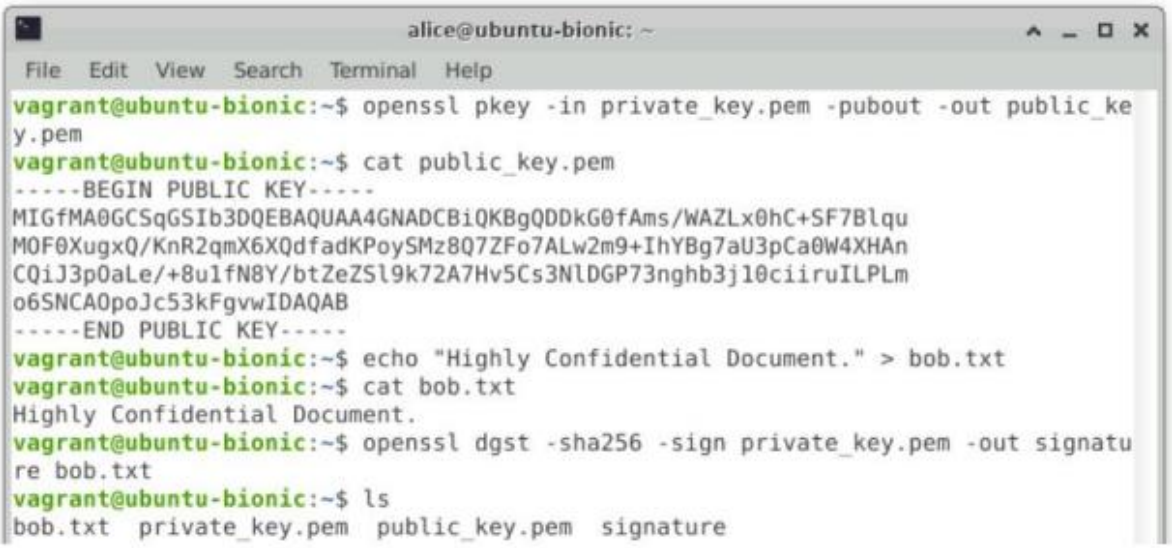
```
vagrant@ubuntu-bionic:~$ sudo adduser bob
Adding user `bob' ...
Adding new group `bob' (1003) ...
Adding new user `bob' (1003) with group `bob' ...
Creating home directory `/home/bob' ...
Copying files from `/etc/skel' ...
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
Changing the user information for bob
Enter the new value, or press ENTER for the default
        Full Name []: bob
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
vagrant@ubuntu-bionic:~$ sudo usermod -a -G sudo bob
```

```
vagrant@ubuntu-bionic:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
```

alice@ubuntu-bionic: ~

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```
alice:x:1002:1002:alice,,,:/home/alice:/bin/bash
bob:x:1003:1003:bob,,,:/home/bob:/bin/bash
vagrant@ubuntu-bionic:~$ openssl genkey -algorithm RSA -out private_key.pem
Invalid command 'genkey'; type "help" for a list.
vagrant@ubuntu-bionic:~$ openssl genpkey -algorithm RSA -out private_key.pem
....+++++
............+++++
vagrant@ubuntu-bionic:~$ cat private_key.pem
-----BEGIN PRIVATE KEY-----
MIICdgIBADANBgkqhkiG9w0BAQEFAASCAmAwggJcAgEAAoGBAMOQbR8Caz9YBkvH
SEL5IXsGWq4w4XRe6DFD8qdHaqZfpdB19p0o+jJIzPxDtkWjsAvDab34iFgGDtpT
ekJrRbhccCcJCInek5ot7/7y7V83xj9u1l5lKX2TvYDse/kKzc2UMY/veeCFvePX
RyKKu4gs8uajpI0IA6mglzneQWC/AgMBAAECgYBrv0sf5qlIQStW3PlHTRPRQHL9
vhi5kQDFA/bRi1uLRu+IpzW06TrQMaQe6P+n5sO793JvJFuag9vzFkMqDQhfz+Kb
3oPBQd6asxK6F78y6Bnzpik5tWI38aQ9bTf5Glln3DA0kl17zS4UzWQo7lbJOMK3
aNco1kRRtBzcDsDnIQJBAPG/fhCbds4RylU2kIyac6ei+MqlnO8rDKAukwJMsOo8
m5XBAfIiEiOx2gagCifJtlhZDqL7Ril9ZjhFv/jbbssCQQDPF+sRcRMjVrOd+2MJ
1bkd3y/W8pCFwyHi8DyojRs/CSPGSaPkp+CvK9Cstu6CpwLA64R8xZ/SjHIkAr/V
QUNdAkA9TYz/Ux0WdoZzg2m2bfhEG1yxrruHruN5yGaZK48R/BAUnhW2XL2IESk
h3Lr0gfucW73v3Z0UPtZ674ZsjNvAkEAtiBu1vP+xMFnTbTbsLaJCsBRrlO8ziaC
Erb8Ya9XGuTK1YPF4z3Q171M09g9giGmxktaRPD4UY66gvj4BHBU8QJAf/80iGhw
RRBv/ke+l03n5KSuM3Xoff7Lxuq/Qa7qhpDlEmkof7YCJdiI3peYMMGxYxVH8WI7
8LsBEh5fDMfaaA==
-----END PRIVATE KEY-----
```


alice@ubuntu-bionic: ~
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vagrant@ubuntu-bionic:~$ openssl pkey -in private_key.pem -pubout -out public_key.pem
vagrant@ubuntu-bionic:~$ cat public_key.pem
-----BEGIN PUBLIC KEY-----
MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQDDkG0fAms/WAZLx0hC+SF7Blqu
MOF0XugxQ/KnR2qmX6XQdfadKPoySMz8Q7ZFo7ALw2m9+IhYBg7aU3pCa0W4XHAn
CQiJ3pOaLe/+8u1fN8Y/btZeZSl9k72A7Hv5Cs3NlDGP73nghb3j10ciiruILPLm
o6SNCAOpoJc53kFgvwIDAQAB
-----END PUBLIC KEY-----
vagrant@ubuntu-bionic:~$ echo "Highly Confidential Document." > bob.txt
vagrant@ubuntu-bionic:~$ cat bob.txt
Highly Confidential Document.
vagrant@ubuntu-bionic:~$ openssl dgst -sha256 -sign private_key.pem -out signature bob.txt
vagrant@ubuntu-bionic:~$ ls
bob.txt private_key.pem public_key.pem signature


vagrant@ubuntu-bionic:~$ cp bob.txt public_key.pem signature /tmp/
vagrant@ubuntu-bionic:~$ su - alice
Password:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
alice@ubuntu-bionic:~$ cp /tmp/bob.txt ~/
alice@ubuntu-bionic:~$
alice@ubuntu-bionic:~$ cp /tmp/public_key.pem ~/
alice@ubuntu-bionic:~$
alice@ubuntu-bionic:~$ cp /tmp/signature ~/
alice@ubuntu-bionic:~$ ls
bob.txt public_key.pem signature
alice@ubuntu-bionic:~$ openssl dgst -sha256 -verify public_key.pem -signature signature bob.txt
Verified OK
alice@ubuntu-bionic:~$ nano bob.txt
alice@ubuntu-bionic:~$ openssl dgst -sha256 -verify public_key.pem -signature signature bob.txt
Verification Failure
alice@ubuntu-bionic:~$

The following packages are for slightly older Linux releases, including Ubuntu 20.04, CentOS 7, and RHEL 8. These are functionally the same as the ones above and will also work on newer operating systems. However, since they are built on legacy operating systems that lack support for some of the newer compiler optimizations, the resulting executables are larger.

- AES Crypt - Desktop & Command-Line (64-bit Intel/AMD) (rpm | deb | tgz)
- AES Crypt - Command-Line only (64-bit Intel/AMD) (rpm | deb | tgz)

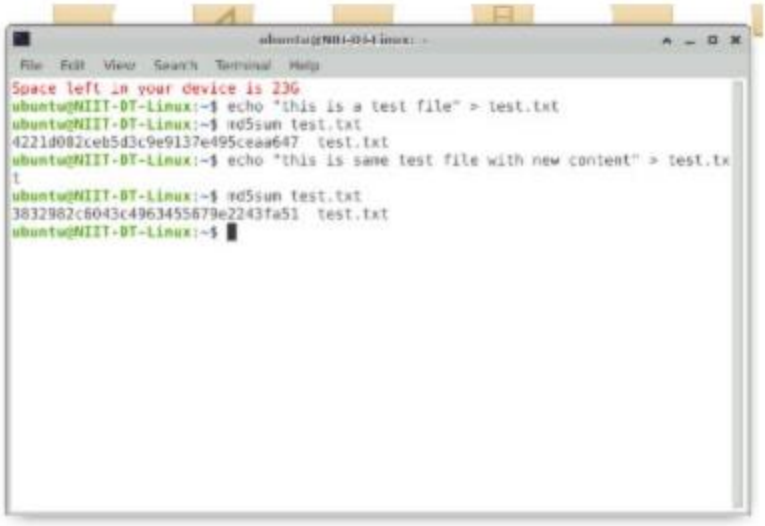
Download files are hashed and the hash file is signed using GnuPG. Additionally, since release 4.3 a separate signature file is available for download. See package verification for details.




```
ubuntu@NIIT-DT-Linux:~/Downloads$ sudo dpkg -i aescrypt_gui-4.5.0-Linux-x86_64.d
eb
(Reading database ... 193790 files and directories currently installed.)
Preparing to unpack aescrypt_gui-4.5.0-Linux-x86_64.deb ...
Unpacking aescrypt_gui (4.5.0) over (4.5.0) ...
Setting up aescrypt_gui (4.5.0) ...
Processing triggers for gnome-menus (3.36.0-1ubuntu1) ...
Processing triggers for desktop-file-utils (0.24-1ubuntu3) ...
Processing triggers for mime-support (3.64ubuntu1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for shared-mime-info (1.15-1) ...
ubuntu@NIIT-DT-Linux:~/Downloads$
```

```
ubuntu@NIIT-DT-Linux:~/Downloads$ aescrypt -e -p P@$$w0rd  smile.jpg
Encrypting: smile.jpg
ubuntu@NIIT-DT-Linux:~/Downloads$
```

```
-DT-Linux:~/Downloads$ rm smile.jpg
ubuntu@NIIT-DT-Linux:~/Downloads$ ls
aescrypt_gui-4.5.0-Linux-x86_64.deb  niitfinal.bmp
code_1.67.2-1652812855_amd64.deb     smile.jpg.aes
ubuntu@NIIT-DT-Linux:~/Downloads$ aescrypt -d -p P@$$w0rd  smile.jpg.aes
Decrypting: smile.jpg.aes
ubuntu@NIIT-DT-Linux:~/Downloads$ ls
aescrypt_gui-4.5.0-Linux-x86_64.deb  niitfinal.bmp  smile.jpg.aes
code_1.67.2-1652812855_amd64.deb     smile.jpg
```


File Edit View Search Terminal Help
Space left in your device is 23G
ubuntu@NIIT-DT-Linux:~$ echo "this is a test file" > test.txt
ubuntu@NIIT-DT-Linux:~$ md5sum test.txt
4221d082ceb5d3c9e9137e495ceaa647 test.txt
ubuntu@NIIT-DT-Linux:~$ echo "this is same test file with new content" > test.tx
t
ubuntu@NIIT-DT-Linux:~$ md5sum test.txt
3832982c6043c4963455679e2243fa51 test.txt
ubuntu@NIIT-DT-Linux:~$